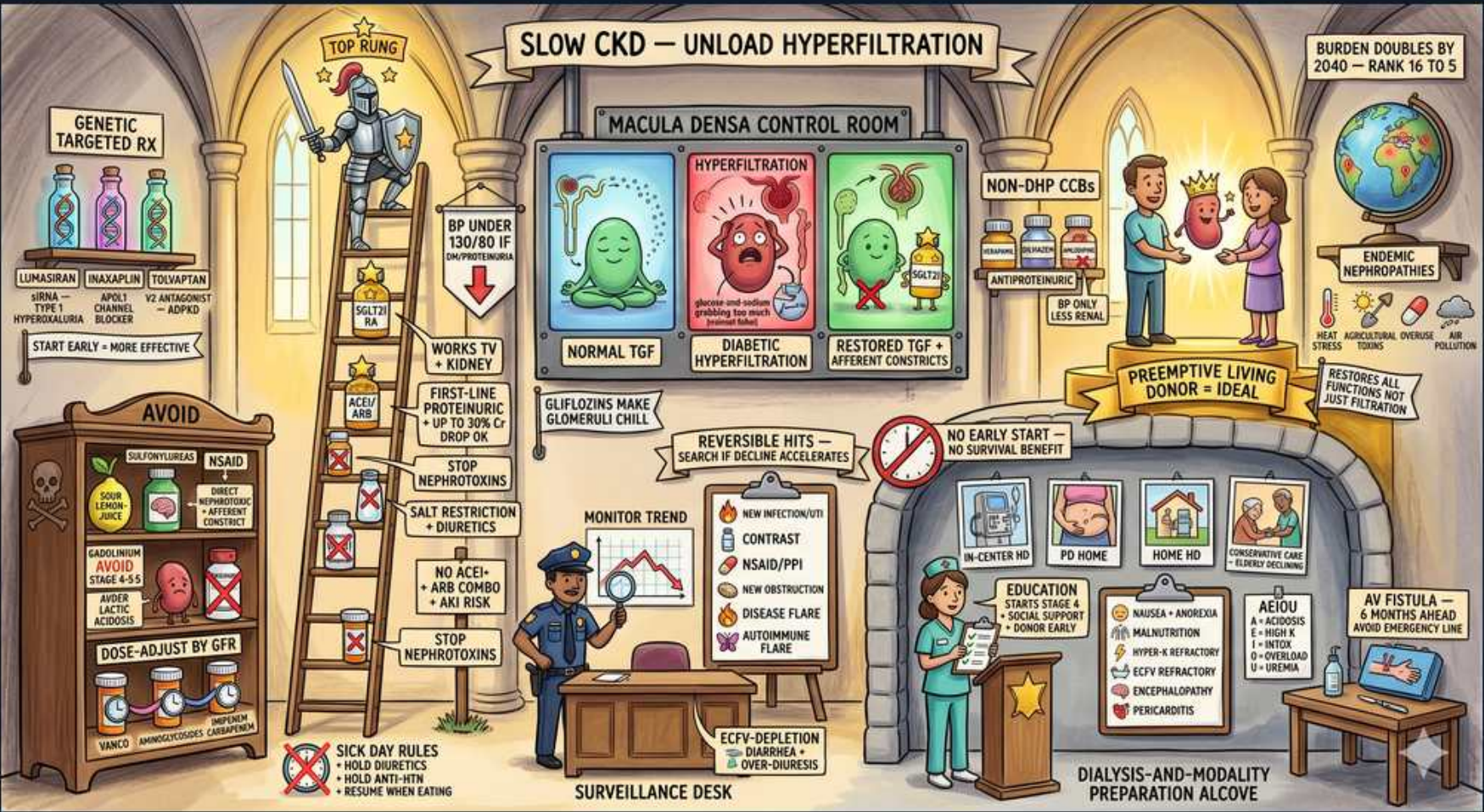


CKD — Management & Slowing Progression



1. Slow CKD — unload hyperfiltration

WHAT YOU SEE

Top banner 'Slow CKD — unload hyperfiltration'.

WHAT IT ENCODES

The unifying goal of CKD therapy is to lower intraglomerular (capillary) pressure and hyperfiltration, which damages surviving nephrons over time. The two evidence-based pillars are RAAS blockade (ACEi/ARB) and SGLT2 inhibitors, layered on BP control <130/80, glycemic control, and lifestyle. Reducing albuminuria is both a treatment target and a marker that hyperfiltration is being unloaded.

BOARD TRAP

WRONG to focus only on lowering creatinine/raising eGFR acutely — both ACEi/ARB and SGLT2i cause an expected initial eGFR dip (up to ~25–30%) from reduced hyperfiltration, which is protective long term, not a reason to stop the drug.

2. Genetic targeted Rx ladder

WHAT YOU SEE

Ladder labeled 'genetic targeted Rx' with named agents.

WHAT IT ENCODES

Whenever a specific cause is identified, start cause-directed therapy early for the greatest benefit: tolvaptan slows ADPKD progression, immunosuppression treats lupus nephritis/ANCA vasculitis, and tight glycemic control plus SGLT2i/finerenone target diabetic kidney disease. Identifying and treating the primary driver complements the generic nephroprotection ladder.

BOARD TRAP

WRONG to apply generic nephroprotection while missing a treatable specific etiology — e.g., giving only an ACEi for rapidly progressive glomerulonephritis instead of urgent immunosuppression, or missing obstruction/lupus. Always pursue the cause; the discriminator is an active urine sediment or rising creatinine that demands a workup.

3. BP under 130/80 knight

WHAT YOU SEE

Knight/shield 'BP under 130/80'.

WHAT IT ENCODES

Blood-pressure target in CKD is generally <130/80 mmHg (KDIGO favors a systolic <120 by standardized measurement in many patients), and the preferred first agents are an ACEi or ARB when albuminuria is present because they lower intraglomerular pressure and proteinuria. Tighter control matters most in proteinuric and diabetic CKD.

BOARD TRAP

WRONG to accept a relaxed <140/90 target for an albuminuric CKD patient — that undertreats proteinuric kidney disease. The discriminator is albuminuria: its presence drives both the tighter BP goal and the choice of an ACEi/ARB as the antihypertensive.

4. Macula densa / TGF panels

WHAT YOU SEE

Three frames 'normal TGF', 'diabetic hyperfiltration', 'restored TGF' (afferent constricts).

WHAT IT ENCODES

In diabetes, proximal SGLT2-driven glucose+sodium reabsorption lowers distal sodium delivery to the macula densa, which is misread as low volume and dilates the afferent arteriole → hyperfiltration. SGLT2 inhibitors (empagliflozin, dapagliflozin, canagliflozin) restore distal sodium sensing, triggering afferent vasoconstriction via tubuloglomerular feedback and dropping intraglomerular pressure.

BOARD TRAP

WRONG to swap the mechanisms: SGLT2i lower glomerular pressure by constricting the AFFERENT arteriole (via tubuloglomerular feedback), whereas ACEi/ARB do it by dilating the EFFERENT arteriole (blocking angiotensin II). Both reduce hyperfiltration but at opposite ends of the glomerulus.

5. Gliflozins make glomerulus chill

WHAT YOU SEE

Caption 'gliflozins make glomerulus chill'.

WHAT IT ENCODES

SGLT2 inhibitors are now first-line nephroprotective agents in CKD — diabetic and (per DAPA-CKD/EMPA-KIDNEY) non-diabetic — reducing albuminuria, slowing eGFR decline, and cutting heart-failure hospitalizations and CV death. They are generally used when eGFR is above ~20–25 mL/min and continued thereafter. Expect a small initial eGFR dip and a modest risk of euglycemic DKA, genital mycotic infections, and volume depletion.

BOARD TRAP

WRONG to stop an SGLT2i because eGFR fell after starting it or because the patient is non-diabetic — the early dip is expected/protective, and the renal/cardiac benefit is independent of diabetes. Do hold it during acute illness (sick-day rule) and watch for euglycemic DKA.

6. Non-DHP CCB cabinet

WHAT YOU SEE

Shelf 'non-DHP CCBs' with king/queen figures.

WHAT IT ENCODES

Non-dihydropyridine calcium channel blockers (diltiazem, verapamil) have an antiproteinuric effect and are useful add-on BP agents in CKD, particularly when an ACEi/ARB is maximized or not tolerated. Dihydropyridine CCBs (amlodipine) lower BP well but are less antiproteinuric and are typically paired with RAAS blockade.

BOARD TRAP

WRONG to combine verapamil/diltiazem with a beta-blocker without caution — non-DHP CCBs depress AV conduction and contractility, risking bradycardia/heart block. They are also contraindicated in HFrEF; the discriminator is their cardiac depressant action versus the vascular-selective DHPs.

7. Preemptive living donor — ideal

WHAT YOU SEE

Banner 'preemptive living donor — ideal'.

WHAT IT ENCODES

The best renal-replacement outcome is a preemptive kidney transplant — ideally from a living donor — performed before dialysis is ever required (generally considered once eGFR <20). It offers the longest patient and graft survival and best quality of life. Early referral for transplant evaluation is key while the patient is still in advanced CKD.

BOARD TRAP

WRONG to require a patient to complete a period of dialysis before transplant — preemptive transplantation is superior and avoids dialysis-associated morbidity. Delaying referral until dialysis-dependent is a missed opportunity, not standard sequencing.

8. No early start / no survival benefit

WHAT YOU SEE

Caption near dialysis prep 'no early start — no survival benefit'.

WHAT IT ENCODES

The IDEAL trial showed that starting dialysis early (eGFR 10–14) gives no survival benefit over starting later (eGFR 5–7) once the patient becomes symptomatic. Dialysis is therefore initiated for clinical indications — uremic symptoms, refractory volume overload, hyperkalemia, or acidosis — not for a number alone.

BOARD TRAP

WRONG to start dialysis purely because eGFR crossed a threshold in an asymptomatic patient. The discriminator is the presence of uremic symptoms or a refractory complication (AEIOU: Acidosis, Electrolytes/hyperK, Intoxications, Overload, Uremia) — these, not the eGFR value, trigger initiation.

9. AVOID cabinet (nephrotoxins)

WHAT YOU SEE

Poison cabinet listing NSAIDs, contrast, metformin/lactic acid cautions.

WHAT IT ENCODES

Avoid nephrotoxins in CKD: NSAIDs (lower GFR via afferent constriction, cause AKI and hyperkalemia), avoidable IV contrast, and aminoglycosides. Metformin risks lactic acidosis and is generally stopped if eGFR <30 and held around contrast; many antibiotics and gabapentinoids accumulate and need renal dosing.

BOARD TRAP

WRONG to give an NSAID for analgesia in CKD 'just short-term' — even brief courses can precipitate AKI and hyperkalemia, especially atop an ACEi/ARB and a diuretic ('triple whammy'). The safer choice is acetaminophen; the discriminator is recognizing the additive prerenal/hemodynamic hit.

10. Salt restriction / stop nephrotoxins

WHAT YOU SEE

List 'salt restriction', 'stop nephrotoxins', protein note.

WHAT IT ENCODES

Dietary sodium restriction (<2 g/day) amplifies the antiproteinuric and BP-lowering effect of ACEi/ARB and SGLT2i. Modest dietary protein moderation (~0.8 g/kg/day) may reduce hyperfiltration, but severe protein restriction is avoided due to malnutrition risk. Stopping ongoing nephrotoxins is part of every CKD plan.

BOARD TRAP

WRONG to push very-low-protein diets to slow CKD — aggressive protein restriction risks protein-energy wasting without clear benefit, and is especially dangerous in elderly/frail patients. The target is moderation (~0.8 g/kg), not starvation of protein.

11. No ACEi + ARB / no MRA combo

WHAT YOU SEE

Caption 'no ACEi + ARB / no MRA' warning.

WHAT IT ENCODES

Do not combine an ACEi with an ARB (dual RAAS blockade): trials (ONTARGET, VA NEPHRON-D) showed more hyperkalemia, AKI and hypotension without added renal/CV benefit. Use one agent, titrated to the maximum tolerated dose. Finerenone (a non-steroidal MRA) can be layered on top of a single RAAS blocker in diabetic CKD with monitoring.

BOARD TRAP

WRONG to add an ARB to an ACEi (or to stack a steroidal MRA carelessly) hoping for more proteinuria reduction — dual blockade is harmful (AKI, hyperkalemia). The discriminator is single-agent RAAS blockade at max tolerated dose; finerenone, not dual ACEi/ARB, is the validated add-on in diabetic CKD.

12. Dose-adjust by GFR

WHAT YOU SEE

Caption 'dose-adjust by GFR'.

WHAT IT ENCODES

Renally cleared drugs accumulate as GFR falls and must be dose-reduced or interval-extended by eGFR — classic examples include many antibiotics (vancomycin, aminoglycosides, beta-lactams), LMWH/enoxaparin, gabapentin/pregabalin, digoxin, and DOACs. Always check renal dosing and avoid contrast/metformin pitfalls.

BOARD TRAP

WRONG to use standard enoxaparin/gabapentin/digoxin doses in advanced CKD — accumulation causes bleeding, sedation/myoclonus, and digoxin toxicity respectively. The discriminator is matching the dose to the current eGFR rather than to the patient's prior baseline.

13. Surveillance desk / monitor trend

WHAT YOU SEE

Officer at desk 'monitor trend' tracking eGFR.

WHAT IT ENCODES

CKD is staged and monitored by eGFR (G1–G5) AND albuminuria (A1–A3, by urine albumin-to-creatinine ratio); both axes independently predict progression and CV risk. Track the trajectory over time — a rapidly declining eGFR or rising ACR signals the need to intensify therapy and refer to nephrology (generally eGFR <30, ACR ≥300, or rapid decline).

BOARD TRAP

WRONG to stage or risk-stratify CKD on eGFR alone — albuminuria is an independent driver of both progression and cardiovascular risk. Two patients at the same eGFR have very different prognoses depending on their ACR (A1 vs A3); ignoring albuminuria understages risk.

14. Sick day rules

WHAT YOU SEE

Caption 'sick day rules' (hold diuretics/ACEi/SGLT2 when ill).

WHAT IT ENCODES

During acute illness with volume depletion (vomiting, diarrhea, sepsis, poor intake), temporarily hold the drugs that impair renal autoregulation or pose metabolic risk: ACEi/ARB, diuretics, SGLT2 inhibitors, metformin, and NSAIDs (mnemonic-style 'SADMANS'). Resume once the patient is rehydrated and stable.

BOARD TRAP

WRONG to keep an SGLT2i running through a febrile, poorly-eating illness — it raises the risk of euglycemic DKA and volume depletion. The discriminator is that these are temporary holds for acute illness, not permanent discontinuations of nephroprotective therapy.

15. Dialysis modality alcove (HD/PD)

WHAT YOU SEE

Nurse with options in-center HD, PD home, home HD; AVF note.

WHAT IT ENCODES

Plan modality and access months ahead: options are in-center hemodialysis, home hemodialysis, and peritoneal dialysis (home-based, more autonomy, peritonitis risk). For HD the preferred access is an arteriovenous fistula, which must be created and allowed to mature ~2–3 months before use; protect that arm from venipuncture/BP cuffs. Order of access preference is AV fistula > AV graft > tunneled catheter (highest infection/thrombosis risk).

BOARD TRAP

WRONG to default to a tunneled central catheter for chronic HD when planning is possible — catheters carry the highest infection and thrombosis risk. The discriminator is timing: refer early so an AV fistula can mature; 'fistula first, catheter last.'
